Lab Report: 5

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Group number: 6

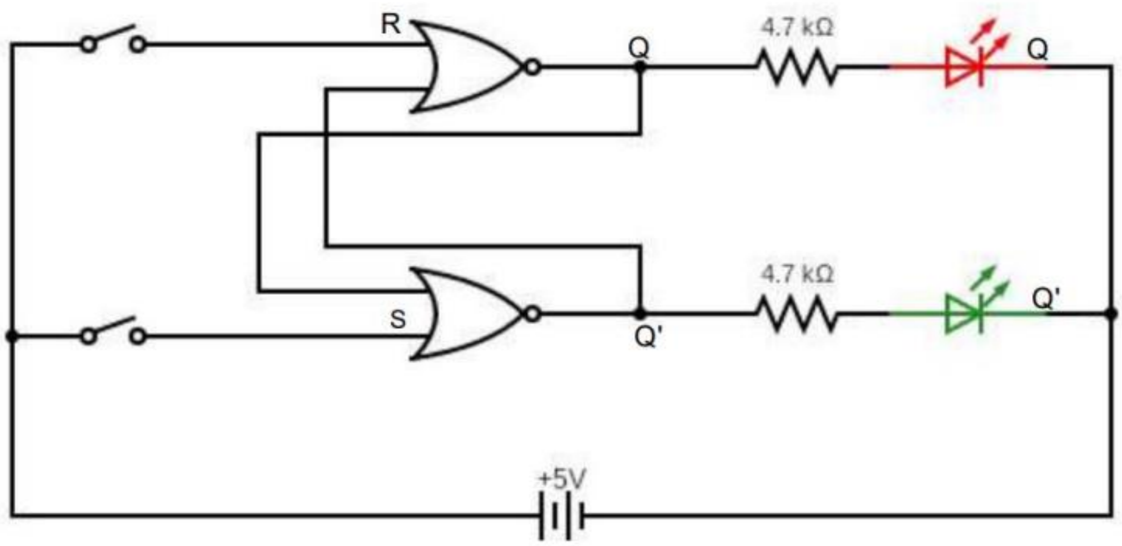
Part-A

Objective: Designing an RS-Latch using NOR gates

Electronic components used:

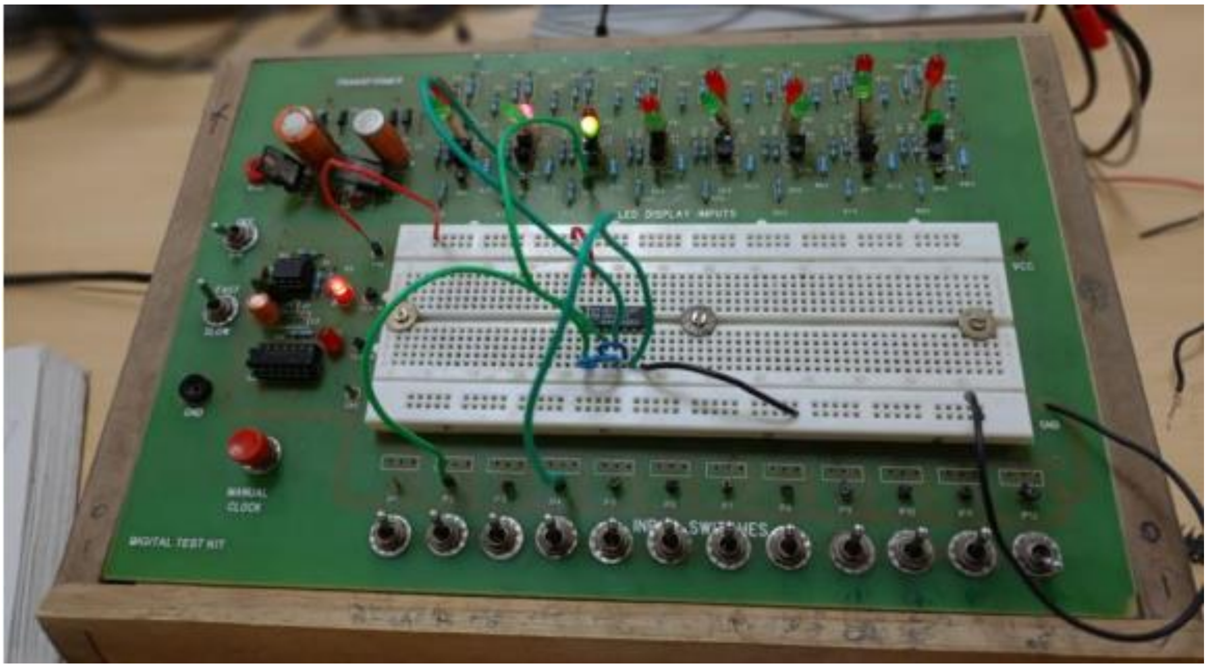
- Digital test kit (w/ breadboard)
- CD4001 (NOR) IC
- Wires

The reference circuit:



Procedure:

- Plug the IC into the breadboard and test it for correct outputs.
- Now, set up the circuit on the breadboard as shown in diagram.
- Power the circuit and provide inputs via the toggle switches. Tabulate the outputs into a state table.
- Provide the following input to the latch circuit in the given order:
    - SR = 01, 00, 10, 00, 01, 10, 01, 00, 11, 00, 10, 11, 00, 01, 11, 00



Conclusion: Given is the output from the SR latch:

S	**R**	**Q_{t+1}**
0	0	Q_t
0	1	0
1	0	1
1	1	Forbidden

Outputs for the sample sequence input:

SR	01	00	10	00	01	10	01	00	11	00	10	11	00	01	11	00
Q	0	0	1	1	0	1	0	0	0	0	1	0	0	0	0	0

Outputs (Q) after SR=11 are indeterministic and can be treated as garbage, since a simultaneous set-reset puts the latch into a random state.

Link for the Tinkercad simulation:

https://www.tinkercad.com/things/8iUQt7w0S5z-parta/editel?sharecode=TM55TRFnMtfMJuWdjfbQIo9cMFi7uVZzH9KMBykBc64

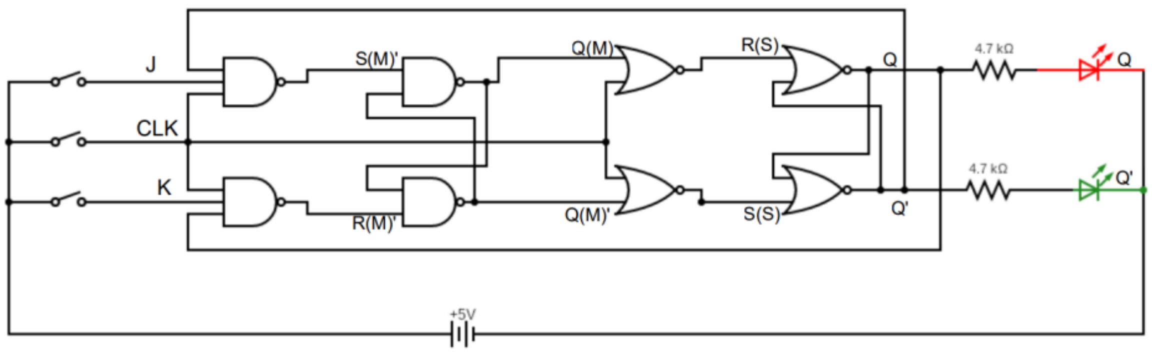
Part-B

Objective: Designing a JK Master-Slave Flip-Flop

Electronic components used:

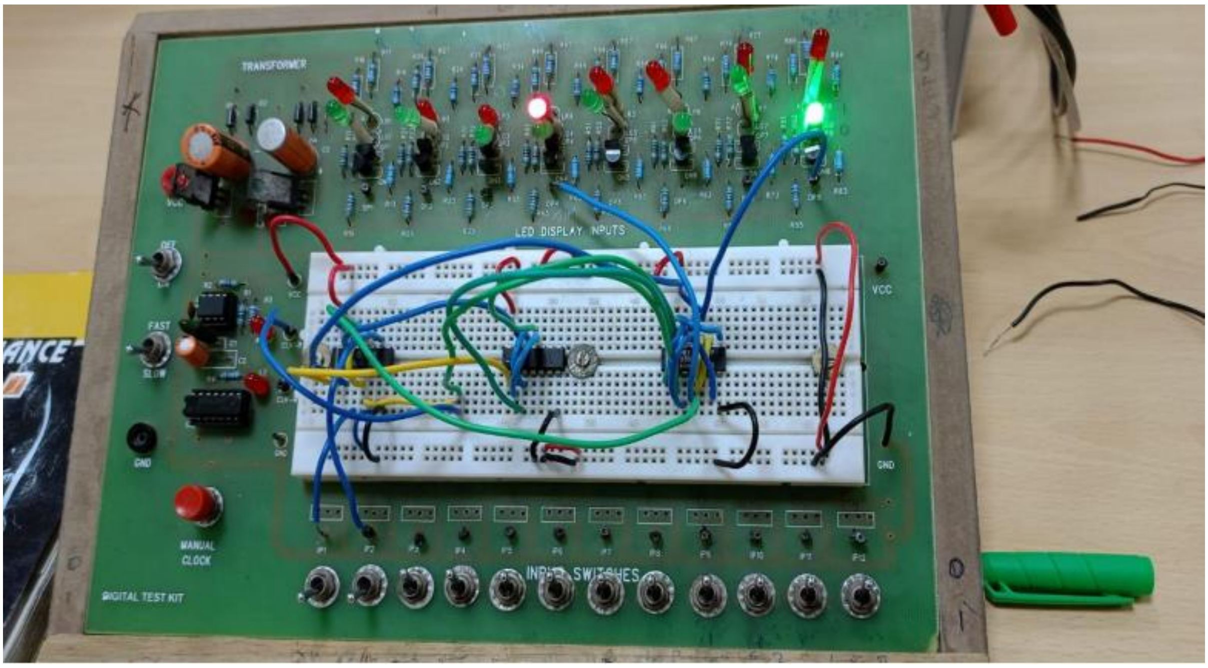
- Digital test kit (w/ breadboard)
- CD4001 (NOR) IC
- 74HC00 (NAND) IC
- CD4012 (quad-input-NAND) ICs
- Wires

The reference circuit:



Procedure:

- Plug the ICs into the breadboard and test them for correct outputs.
- Now, set up the circuit on the breadboard as shown in diagram.
- Power the circuit and provide inputs via the toggle switches.
- Tabulate the outputs against inputs into a state table.



Conclusion: Given is the output from the JK flip-flop:

J	K	Q_{t+1}
0	0	Q_t
0	1	0
1	0	1
1	1	Q'_t

Link for the Tinkercad simulation:

https://www.tinkercad.com/things/h1qpl3RR0uQ-incredible-juttuli-amur/editel?sharecode=wOnDXSd6yFy7FUWcp3xev0nJ-FgRXEGahmm5Rqw4hvk

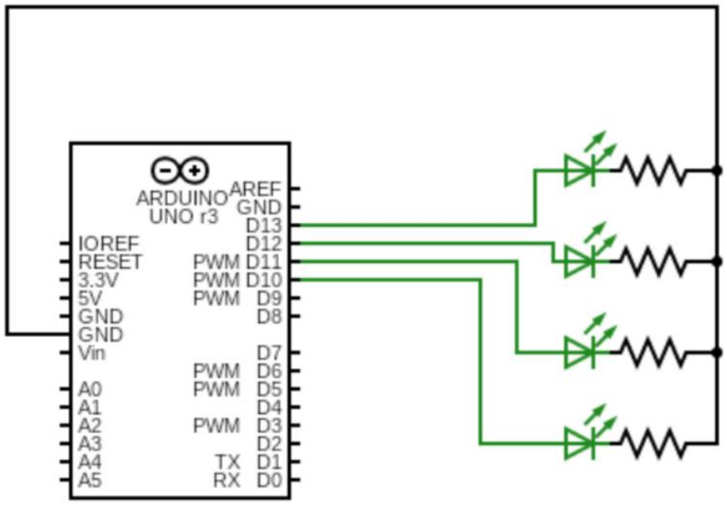
Part-C

Objective: Designing a 4-bit binary up-down ripple counter

Electronic components used:

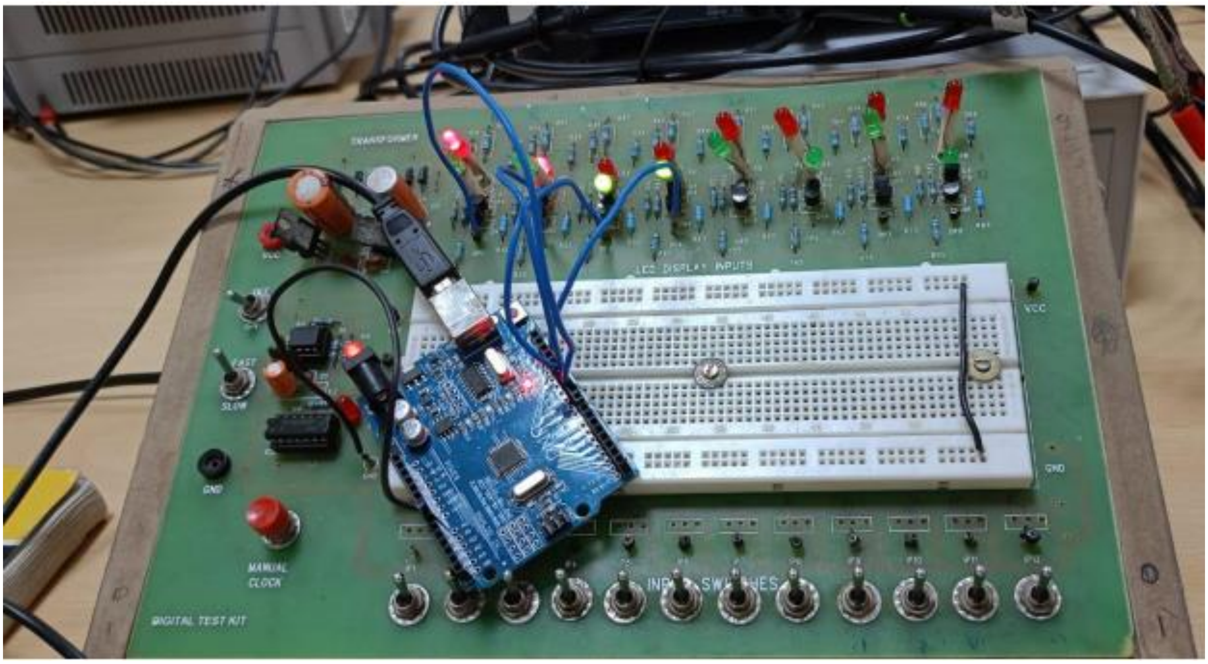
- Digital test kit
- Arduino uno
- Wires

The reference circuit:



Procedure:

- Connect the ground from Arduino to the ground of the digital kit.
- Provide inputs to the LEDs using the Arduino pins.
- Plug in the USB to the Arduino and select the appropriate board and port in the IDLE.
- Verify and upload the code (as described in the manual).



Conclusion: The ripple counter counts up from 0 to 15, then decays back to 0, and repeats, as intended.

Link for the Tinkercad simulation:

https://www.tinkercad.com/things/lfKAUBh4JUF-smooth-fyyran/editel?sharecode=C3KHJsb9utOpZVryxQx8oqS24ZzDC_DWA7BDkoJezgQ